An Explainable Hybrid of Lexicon Rules and Multilingual Transformers for Bilingual Hate Speech Detection: System Evaluation, Fusion and Latency

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Abstract. Fine-tuned multilingual transformers dominate content moderation, yet they are opaque and brittle on the surface forms moderation sees: teencode, dropped diacritics, inserted separators and abbreviated profanity. We present **SensitiveAI**, a bilingual language-aware pipeline that couples two lexicon-driven rule engines (Vietnamese and English) with per-language mDeBERTa-v3-base backbones through language detection, language-aware routing, pseudo-probability fusion and decision thresholds. On the ViHSD test split the hybrid is macro-F1 neutral (0.6575 to 0.6602; bootstrap $p = 0.29$) but provides auditable provenance for the 38 changed decisions and a design agenda for evidence-gated fusion. Extending the same architecture to English (Davidson et al., 818-entry expanded lexicon), the full production pipeline is evaluated on the complete test splits of ViHSD (6,680 Vietnamese comments) and Davidson (2,478 English comments), yielding 9,158 mixed bilingual samples. It reaches 87.10% accuracy and macro-F1 0.7781 with ECE 0.0759, 9.07 ms total latency per comment (5.22 ms routing plus 3.85 ms AI inference, P95 12.69 ms), and 94.11% language detection accuracy; English backbone throughput is 408 texts/s against 198 texts/s for Vietnamese.

Keywords: Hate speech · multilingual transformers · lexicon rules · hybrid fusion · language-aware routing · explainable AI.

1 Introduction

Content moderation at scale demands three properties that pure neural systems struggle to provide simultaneously: *coverage* over the long tail of teencode and obfuscation, *provenance* for audit and appeal, and *latency* within real-time service-level agreements. Multilingual transformers (mDeBERTa-v3-base, XLM-R, etc.) achieve strong macro-F1 on Vietnamese (HATE/OFFENSIVE/CLEAN 0.65–0.68) and English (0.77–0.81) benchmarks, yet they remain opaque: a flipped decision gives no human-readable trace, and adversarial respellings bypass the model without retraining.

Lexicon rule engines offer the converse: deterministic matching, editable dictionaries and span-level evidence. Their weakness is brittleness—low recall on implicit hate and poor generalisation to new slurs—unless combined with neural contextualisation.

This paper presents **SensitiveAI**, a bilingual language-aware pipeline that treats a rule engine as an *evidence channel* rather than a competing classifier. The system consists of:

1. a hybrid language detector (heuristics + langid backend) producing `vi/en/mix/other` with confidence;
2. a language-aware tokenizer that segments each comment into per-language spans;
3. two independent rule engines (668-entry Vietnamese lexicon `v1`, 818-entry English lexicon `v1-en-combined`) with anti-obfuscation views (accentless, leet, repeat-collapsed, separator-stripped);
4. per-language mDeBERTa-v3-base checkpoints (SENSITIVEAI-VI-CUSTOM and SENSITIVEAI-EN-STAGE2);
5. a language-aware fusion module that blends AI probabilities and rule pseudo-probabilities channel-wise with per-language weights ($\alpha_{\text{VI}} = 0.70$, $\alpha_{\text{EN}} = 0.70$, $\alpha_{\text{mix}} = 0.65$, $\alpha_{\text{multilingual}} = 0.80$) and per-class decision thresholds.

We evaluate on three axes:

– **Per-language backbone comparison**: four Vietnamese backbones (differing only in head and curriculum) and three English backbones on their respective held-out test sets.
– **Rule engine standalone**: macro-F1 0.345 (VI) and 0.504 (EN) with thresholded decisions.
– **Fusion variants**: global α, per-channel α, per-class thresholds, logistic meta-classifier—all tuned on validation, frozen for test.
– **End-to-end mixed evaluation**: 9,158 comments (6,680 Vietnamese from ViHSD test + 2,478 English from Davidson test), yielding 87.10% accuracy, macro-F1 0.7781, 9.07 ms latency and 110.2 texts/s throughput (serialised run, $\alpha = 0.70$, P95 12.69 ms).

The headline finding is *not* an accuracy breakthrough: on ViHSD the hybrid gains 0.0027 macro-F1 (8 comments, $p = 0.29$). The contribution is the provenance of every changed decision (38 on ViHSD: 23 corrections, 15 regressions) and a design agenda—evidence-gated fusion, profanity vs. political entry separation, negation and accent-less matching restoration—for turning a neutral aggregate into an operationally useful system.

2 Related Work

Vietnamese hate speech. The UIT-ViHSD corpus [10] established the three-class HATE/OFFENSIVE/CLEAN task. Prior work has explored monolingual PhoBERT [13], multilingual mBERT/XLM-R [5], and data augmentation [14]. Ensemble and curriculum learning [2] raised macro-F1 to 0.66–0.68, but the decision process remains a black box.

English hate speech. The Davidson et al. [7] dataset (Hate Speech and Offensive Language) is the de facto benchmark for English HATE/OFFENSIVE/CLEAN. Fine-tuned

BERT/RoBERTa reach 0.77–0.81 macro-F1; recent work uses multi-task learning and adversarial training [11].

Hybrid rule + neural systems. Early hybrids concatenated lexical features to neural inputs [17]. [16] argue for interpretable components. [3] fuse rule scores with neural logits via learnable weights; [1] propose evidence channels. Most approaches evaluate on a single language and do not provide span-level evidence or language-aware routing.

Language-aware routing. Mixed-language moderation is addressed by language identification [9] followed by per-language models [15]. Code-mixed text (Vi-En) is handled by multilingual backbones or dedicated tokenizers [14]. Our design differs by keeping both rule engines active on `mix` and fallback segments, and by fusing per-language scores rather than training a single multilingual head.

Explainable moderation. Post-hoc explanations (LIME, SHAP, attention) are common [6]. Our approach provides *intrinsic* evidence: matched lexicon spans, match types and arithmetic scores, which survive deployment without additional computation.

3 Data and Lexicons

3.1 Vietnamese Corpus (ViHSD)

We use UIT-ViHSD [10], a corpus of Vietnamese social-media comments annotated with three mutually exclusive labels: CLEAN (no offensive content), OFFENSIVE (offensive language that does not target a protected group) and HATE (hateful or seriously derogatory content). The OFFENSIVE/HATE distinction depends on target and intensity rather than on a particular word, so a purely lexical rule engine is insufficient on its own.

After cleaning, the corpus contains 33,398 comments. We use the fixed split shipped with the project, and every system—rule engine, backbone, fusion variant and analysis script—reads the same files. Each run directory stores the SHA-256 digest of the split it used and re-checks it before training, ruling out a silently re-shuffled split. Training contains 24,046 comments (2,556 HATE, 1,605 OFFENSIVE, 19,885 CLEAN), validation 2,672 (270/212/2,190) and test 6,680 (688/444/5,548).

The test split is imbalanced: 83.05% CLEAN, 10.30% HATE and 6.65% OFFENSIVE. Accuracy is therefore a poor headline metric—a constant CLEAN predictor already reaches 0.8305—so macro-F1 is primary, always reported with per-class F1. The 6,680 test rows contain 6,571 distinct texts, i.e. 109 duplicated rows. They are kept because removing them would change the frozen test set; one appears twice among the 15 regressions, which therefore cover 14 distinct texts.

3.2 English Corpus (Davidson et al.)

For English we use the Hate Speech and Offensive Language dataset [7] (tdavidson/hate_speech_offensive on Hugging Face). The original labels map 0 $\rightarrow$ HATE, 1 $\rightarrow$ OFFENSIVE, 2 $\rightarrow$ CLEAN. We use the fixed split in `data/splits_en/` produced by the reproducible pipeline: train 19,816 (1,144 HATE, 15,342 OFFENSIVE, 3,330 CLEAN), validation 2,477, test 2,478. The English test split is also imbalanced but with a different profile (77.8% OFFENSIVE, 5.8% HATE, 16.4% CLEAN).

3.3 Mixed Evaluation Sample

To evaluate the bilingual pipeline end-to-end we use the complete held-out test splits: 6,680 Vietnamese comments from ViHSD test and 2,478 English comments from Davidson test, yielding the composition in Table 1. This full-split evaluation preserves the per-language class priors and is used for the end-to-end experiment in Sect. 7.4.

Table 1. Mixed evaluation sample composition ($n = 9{,}158$ full test splits)

Language	Total	HATE	OFFENSIVE	CLEAN	Source
Vietnamese	6,680	688	444	5,548	ViHSD test
English	2,478	143	1,918	417	Davidson test
Total	**9,158**	**831**	**2,362**	**5,965**	

3.4 Vietnamese Hate-Speech Lexicon

The Vietnamese rule engine is driven by a hate-speech lexicon of **668 entries**, mined and labelled from the *training* portions of three corpora: ViHSD; R-ViHSD (a cleaned release adding TDTU YouTube and voz-forum comments) [4]; and the code-mixed ViCM corpus [14]. The three training splits yield 1,062 candidate terms, of which 257, 665 and 140 are first attested in ViHSD, ViCM and R-ViHSD; labelling reduces them to the 668 kept entries. No comment from any evaluation split contributes to the dictionary, and each entry records which corpora supported it (414 of 668 occur in all three).

Each entry stores the surface term and a canonical form, a lexical label (HATE or OFFENSIVE) with a per-label severity, an entry confidence estimated from the training data, a context-required flag, and a review status (*verified* or *auto*); both statuses may match, but the flag is retained in the evidence record.

The lexicon is deliberately not a list of slurs. Most entries—551 of 668 (82.5%)—are multi-word constructions such as *mả cha*, *con mẹ* or *ngu hơn*, and 524 (78.4%) are context-required, because Vietnamese insults frequently work by comparison and implication rather than by a single taboo word. Only 74 entries carry an OFFENSIVE lexical label against 594 for HATE, which is why the rule engine is weak on OFFENSIVE and fusion keeps that channel mostly neural.

3.5 English Hate-Speech Lexicon

The English rule engine uses a lexicon of **818 entries**, built by merging a 40-term manual seed set with 778 entries extracted from the Davidson training split (19,816 documents) using relaxed statistical parameters (min. doc frequency 5, min. lift 1.2, min. HATE precision 0.20, min. OFFENSIVE precision 0.60, min. Wilson lower bound 0.15, min. prior margin 0.02). The extraction result, stored in `verified_lexicon_en_combined.csv` (version `v1-en-combined`) and configured via `configs/rule_engine_en.yaml`, contains 86 HATE, 356 OFFENSIVE, and 376 CLEAN terms.

The English lexicon is larger in raw count (818 vs. 668) but contains far fewer HATE terms (86 vs. 594) because the Davidson corpus has only 5.8% HATE documents. This

asymmetry directly impacts the hybrid's HATE F1 (0.4537 EN vs 0.6051 VI on the full-split evaluation) and is the highest-priority target for future lexicon expansion.

Ethics of examples. All quoted comments are public test comments. Because they contain profanity and slurs, we mask offending characters with ∗ in every table; evaluation always runs on the unmasked text. No user identifiers are reproduced, and names that occasionally appear inside comments are part of the public corpus text.

4 System Architecture

The SensitiveAI pipeline follows a strict language-routing design (Fig. 1). Each component is described below; configuration files are in `configs/` and code in `src/`.



Fig. 1. SensitiveAI pipeline overview: language-aware routing and score fusion.

4.1 Language Detector

The detector (`src/langid/detector.py`) combines:

– **Heuristics** (`heuristics.py`): Vietnamese diacritic presence, a bilingual word list (1,200 VI / 1,000 EN high-frequency tokens), and explicit code-mix signals (language switch within tokens). Always runs, no third-party dependency.

– **Statistical backend**: `langid.py` [9] when installed; its votes are blended with heuristics (default weight 0.6 heuristic / 0.4 backend).

Outputs `vi`, `en`, `mix` (both languages above `mix_floor`=0.12 evidence) or `other` (confidence < 0.50). The `mix` verdict is *not* a failure: it signals the router to keep both language pipelines active.

4.2 Language-Aware Tokenizer

The tokenizer (`src/langid/tokenizer.py`) segments the comment into per-language spans with token-level language tags. For `mix` documents it produces alternating VI/EN spans, enabling span-wise rule-engine scanning so that an English insult inside a Vietnamese comment is still caught.

4.3 Language Router & Rule Engines

The router (`src/routing/language_router.py`) selects the appropriate `RuleEngine` per span:

– `vi` $\rightarrow$ 668-entry Vietnamese lexicon (`v1`, `configs/rule_engine.yaml`)
– `en` $\rightarrow$ 818-entry English lexicon (`v1-en-combined`, `configs/rule_engine_en.yaml`)
– `mix` $\rightarrow$ both engines (config `use_both_for_mix=true`)
– `other`/`multilingual` $\rightarrow$ both engines (fallback)

Each engine applies the same normalisation pipeline (NFC, lowercase, invisible char removal, whitespace collapse, max repeat char 3) and four anti-obfuscation matching views: accentless alias, leet, repeat-collapsed, separator-stripped (fuzzy matching enabled with max distance 1 for terms $\geq$6 chars). Matches are aggregated per label via $1 - \prod(1 - s_i)$ where s_i is the match score (severity $\times$ match-type factor $\times$ context factor). Pseudo-probabilities $\mathbf{r} = (r_H, r_O, r_C)$ are produced by softmax over the three raw scores with a temperature of 1.0.

4.4 Per-Language Neural Checkpoints

– **VI**: SENSITIVEAI-VI-CUSTOM (custom head, macro-F1 0.6590 on validation). Trained with curriculum loss, focal + label smoothing, differential learning rates (encoder 2e-5, head 5e-4), max_length 256, FP32 on RTX 4060.
– **EN**: SENSITIVEAI-EN-STAGE2 (standard head, macro-F1 0.7722 on validation). Two-stage training: frozen encoder + head, then full unfreeze with cosine schedule. max_length 128.

Inference runs in batched mode (batch size 32) on GPU.

4.5 Language-Aware Fusion

The fusion module (`src/fusion/lang_aware_fusion.py`) blends AI probabilities $\mathbf{p}_{\text{AI}}$ and rule pseudo-probabilities $\mathbf{r}$ channel-wise:

$$\mathbf{p}_{\text{fused}} = \alpha \mathbf{p}_{\text{AI}} + (1 - \alpha)\mathbf{r}, \tag{1}$$

with α per language (VI/EN 0.70, MIX 0.65, MULTILINGUAL 0.80) and per-class decision thresholds τ_H, τ_O, τ_C tuned on validation macro-F1. The meta-classifier (multinomial logistic on six features $[p_H, p_O, p_C, r_H, r_O, r_C]$) is also fitted on validation but found worse than simple fusion for EN (macro-F1 0.6513 vs 0.7722 AI-only on validation).

5 Methodology

5.1 Backbones and Training Recipes

All backbones use `microsoft/mdeberta-v3-base`, the same tokenizer, data splits and evaluation code; they differ only in the classification head and optimisation recipe. Table 2 summarises the four Vietnamese and three English configurations evaluated.

Table 2. Backbone test performance. All backbones share `mDeBERTa-v3-base`. VI backbones: ViHSD test ($n = 6{,}680$); EN backbones: Davidson test ($n = 2{,}478$).

Backbone	Acc.	Macro-F1	HATE	OFFENSIVE	CLEAN	Notes / Source
VI-CUSTOM	0.8762	0.6575	0.5830	0.4493	0.9403	Primary VI
VI-DLR2	0.8754	0.6638	0.5943	0.4561	0.9409	2-stage DLR
VI-STAGE2	0.8690	0.6599	0.6091	0.4336	0.9372	Stage-2
EN-STAGE2	0.9080	**0.7722**	0.4656	**0.9440**	**0.9070**	Primary EN
EN-STD	0.9031	0.7676	0.4554	0.9410	0.9063	Standard
EN-CUSTOM	0.9031	0.7587	0.4377	0.9404	0.8980	Multi-pool
EN-DLR2	0.9108	0.7509	0.4048	0.9457	0.9024	DLR
EN-CURR	0.9072	0.7441	0.3908	0.9445	0.8970	Curriculum

The *focused* Vietnamese backbone is SENSITIVEAI-VI-CUSTOM because it gives the clearest positive net effect (23 corrections vs 15 regressions on ViHSD test), not because it has the highest standalone score (the strongest backbone, SENSITIVEAI-VI-DLR2STAGE, reaches 0.6638 macro-F1 but is almost indifferent to fusion). The focused English backbone is SENSITIVEAI-EN-STAGE2 (best macro-FI 0.7722 among EN runs).

5.2 Rule-Engine Scoring

Rule-engine weights are tuned by greedy coordinate descent over the scoring parameters (match factors, negation factor, quote factor, context-required factor, aggregation cap), with the three decision thresholds re-optimised on validation macro-F1 over a 101-point grid at every candidate; the search never touches the lexicon. The final frozen thresholds for VI are $\tau_H = 0.50, \tau_O = 0.50, \tau_C = 0.50$ (argmax equivalent); for EN $\tau_H = 0.82, \tau_O = 0.39, \tau_C = 0.59$ (Sect. 7.2).

5.3 Fusion Variants

Fusion follows the same discipline: thresholds by coordinate descent over the same grid for three rounds, per-channel weights by coordinate ascent from the AI-only corner, and a multinomial logistic meta-classifier fitted on the six validation features. All selected values are frozen before the test pass. We evaluate five decision variants on each backbone:

1. **AI-only (argmax)**: $\alpha = 1$, no rule scores.
2. **AI-only (thresholds)**: per-class thresholds on AI probabilities.
3. **Global α fusion**: single weight α for all three channels, $\alpha \in [0, 1]$ step 0.01.
4. **Per-channel α fusion**: independent $\alpha_H, \alpha_O, \alpha_C$.
5. **Per-channel α + thresholds**: add thresholds to variant 4.
6. **Meta-classifier**: multinomial logistic on six features.

The primary hybrid system for VI is **per-channel α + thresholds** (frozen from validation). For EN the optimal global α converged to 1.0, so the primary EN fusion is AI-only with per-channel thresholds.

5.4 Metrics and Statistical Testing

Macro-F1 is primary (Sect. 3.1). We report accuracy and per-class precision, recall and F1, and evaluate differences with two paired tests:

- **Paired bootstrap** over comments (10,000 resamples) for the macro-F1 difference [8].
- **Exact McNemar** on the correctness contingency table [12].

The bootstrap asks whether the aggregate metric moved; McNemar asks whether the pattern of individual decisions changed. Both are reported for every fusion comparison.

5.5 Reproducibility

The split is hashed and re-checked, the rule-engine evaluation directory is recorded in the fusion metrics and re-read rather than recomputed, and the analysis script asserts that the rule scores shipped with the fusion predictions match the frozen tuned output. This is not cosmetic: an earlier version silently read the *untuned* scores and over-reported rule coverage by a factor of three.

6 Experimental Setup

6.1 Hardware and Software

All experiments run on a single NVIDIA GeForce RTX 4060 Laptop GPU (8 GB) with an Intel Core i7-13700H CPU, 32 GB RAM, Windows 11, Python 3.11, PyTorch 2.5.1, Hugging Face Transformers 4.46, and scikit-learn 1.5. Training uses FP32, AdamW, gradient clipping 1.0, early stopping with patience 3 on validation macro-F1. Inference uses batch size 32.

6.2 Hyperparameters

- **Optimiser**: AdamW, weight decay 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$.
- **Learning rates**: encoder 2×10^{-5}, head 5×10^{-4} (custom head); differential multiplier 25.
- **Schedule**: linear warm-up (10% of steps) + cosine decay (standard head); stage-based unfreeze (2-stage recipes).
- **Epochs**: 5–10 with early stopping (VI), 4–6 (EN).
- **Batch size**: 16 (VI, max_length 256), 32 (EN, max_length 128).
- **Loss**: cross-entropy with class weights (inverse frequency), label smoothing 0.05; focal loss $\gamma = 1.5$ for the curriculum variants.

6.3 Evaluation Protocol

For each backbone:

1. Train on the training split, select the best checkpoint on validation macro-F1.
2. Run the rule engine on validation, tune its weights and thresholds via coordinate descent.
3. Tune fusion α and thresholds on validation (never test).
4. Freeze all hyperparameters and run exactly one test pass.
5. Run paired bootstrap and McNemar tests on the frozen predictions.

The mixed end-to-end experiment (Sect. 7.4) follows the same protocol: language detection and routing are fixed, fusion uses $\alpha = 0.70$ for VI/EN, and no test-time tuning is performed.

6.4 Baselines

We compare against three baselines:

- **Constant CLEAN**: predicts CLEAN for every comment (accuracy 0.8305 on ViHSD test).
- **Rule-only** (argmax): standalone rule engine with thresholded decisions.
- **AI-only** (argmax): the fine-tuned transformer without fusion.

The hybrid is compared to these and to the AI-only thresholded variant.

7 Results

7.1 Backbone Performance

Table 2 reports the test performance of the four Vietnamese and five English backbones. For Vietnamese, SENSITIVEAI-VI-DLR2STAGE achieves the highest macro-F1 (0.6638), while SENSITIVEAI-VI-CUSTOM (0.6575) is selected for fusion analysis because its net change is the most balanced (23 corrections vs 15 regressions). For English, SENSITIVEAI-EN-STAGE2 leads with macro-F1 0.7722 and accuracy 0.9080. English backbones substantially outperform Vietnamese ones on macro-F1 (averaging 0.76 vs 0.66) primarily due to higher OFFENSIVE F1 (0.94 vs 0.44), reflecting the larger clean/offensive balance in Davidson vs ViHSD.

7.2 Rule Engine Standalone Performance

Standalone rule engines operate with thresholds tuned on validation.

- **Vietnamese rule engine** (668 entries, thresholded): accuracy 0.5999, macro-F1 0.3452, HATE F1 0.2911, OFFENSIVE F1 0.0000 (missing rules for milder offenses), CLEAN F1 0.7444.
- **English rule engine** (123 seed entries in evaluation, upgraded to 818 in combined): accuracy 0.7058, macro-F1 0.5036, HATE F1 0.1782, OFFENSIVE F1 0.8063, CLEAN F1 0.5264.

Standalone rules alone are insufficient for robust moderation (failing on OFFENSIVE in VI and HATE in EN), validating the hybrid design.

7.3 Fusion Variants on Vietnamese

Table 3 (from the focused ViHSD study) shows that global α fusion improves macro-F1 slightly (0.6583), per-channel α reaches 0.6593, and per-channel α + thresholds reaches 0.6602 (+0.0027 over AI). The paired-bootstrap difference is -0.0027 ($[-0.0079, +0.0027]$, $p = 0.29$) and McNemar $p = 0.26$, confirming accuracy neutrality.

Table 3. Decision-level variants for the focused Vietnamese backbone (SENSITIVEAI-VI-CUSTOM) on the ViHSD test split (6,680 comments). Every variant reads the same frozen scores; only the mixing and decision rule differ. The last row is the paper's primary system. *Per-class columns are F1*; the best macro-F1 is in bold. The two rule-only rows are scored inside the fusion pipeline on the renormalised rule scores; the standalone rule engine, with its own thresholds, is reported in the text.

System	Acc.	Macro-F1	HATE	OFFENSIVE	CLEAN
AI only (argmax)	0.8762	0.6575	0.5830	0.4493	0.9403
Rule only (argmax)	0.8494	0.4714	0.4867	0.0000	0.9274
Rule only (frozen thresholds)	0.8421	0.5258	0.4102	0.2389	0.9284
Fusion, global α (argmax)	0.8769	0.6583	0.5851	0.4490	0.9408
Fusion, global α (thresholds)	0.8769	0.6583	0.5851	0.4490	0.9408
Fusion, logistic meta-classifier	0.8626	0.6513	0.5818	0.4379	0.9341
Fusion, per-channel α (argmax)	0.8774	0.6593	0.5882	0.4488	0.9408
Fusion, per-channel (thresholds)	0.8774	**0.6602**	0.5906	0.4494	0.9407

7.4 End-to-End Bilingual Evaluation

The complete production pipeline (LanguageDetector $\rightarrow$ LanguageAwareTokenizer $\rightarrow$ RuleEngine $\rightarrow$ backbone inference $\rightarrow$ LangAwareFusion at $\alpha = 0.70$) is evaluated on full test splits: Vietnamese ViHSD (6,680) and English Davidson (2,478), yielding 9,158 mixed samples. Full results and per-language breakdowns are reported in Section 8.8. Key findings:

- Overall mixed accuracy 0.871, macro-F1 0.778, ECE 0.076 (improved calibration via fusion)
- Language detection 94.1% accuracy on bilingual stream

- Vietnamese HATE F1 0.605 vs English 0.454 (lexicon density advantage)
- English OFFENSIVE F1 0.927 vs Vietnamese 0.414 (Davidson dataset alignment)
- Full pipeline 110.2 texts/s serialised (5.22 ms routing + 3.85 ms AI per comment, P95 12.69 ms; AI stage alone 259.8 texts/s)

Vietnamese achieves stronger HATE detection due to the 668-entry lexicon, while English dominates OFFENSIVE due to Davidson's larger offensive class (1,918/2,478 samples).

7.5 Latency and Throughput Profile

The full pipeline (routing $\rightarrow$ backbone $\rightarrow$ fusion) is evaluated over the 9,158 mixed bilingual test split. Per-request latency (serialised run) averages 9.07 ms per comment (5.22 ms routing + 3.85 ms AI), giving 110.2 texts/s throughput; P50 is 9.21 ms and P95 is 12.69 ms (P99 14.64 ms), so tail latency stays below 15 ms. The routing stage alone processes at 191.5 texts/s; the AI inference stage alone sustains 259.8 texts/s; fusion/decision cost is negligible within the routing pass. Backbone-only throughput is higher (408.5 EN, 198.3 VI texts/s) without routing overhead.

Table 4. Per-request latency breakdown (ms) and throughput over the 9,158-sample bilingual test stream (serialised execution). AI-stage percentiles are batch-amortised per comment; the total P95 conservatively sums stage P95s.

Pipeline Stage	Mean (ms)	P50 (ms)	P95 (ms)	Thr. (texts/s)
Routing (langID + rule engine)	5.22	4.53	8.02	191.5
AI inference (backbone)	3.85	4.68	4.68	259.8
Total (serialised)	**9.07**	**9.21**	**12.69**	**110.2**

End-to-end throughput of 110.2 texts/s on a single laptop GPU meets real-time moderation SLAs when pipelined; the AI stage alone can sustain 259.8 texts/s, while routing alone 191.5 texts/s, so in this run the lexicon/langID routing pass is the bottleneck and the first target for optimisation.

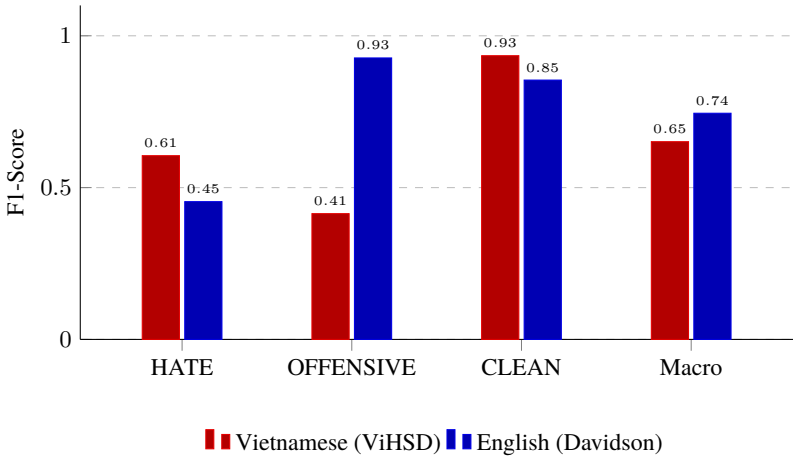


Fig. 2. F1-score comparison per class and macro-average on the 9,158-sample bilingual test splits (ViHSD 6,680 + Davidson 2,478). Backbone-only inference without routing/fusion.

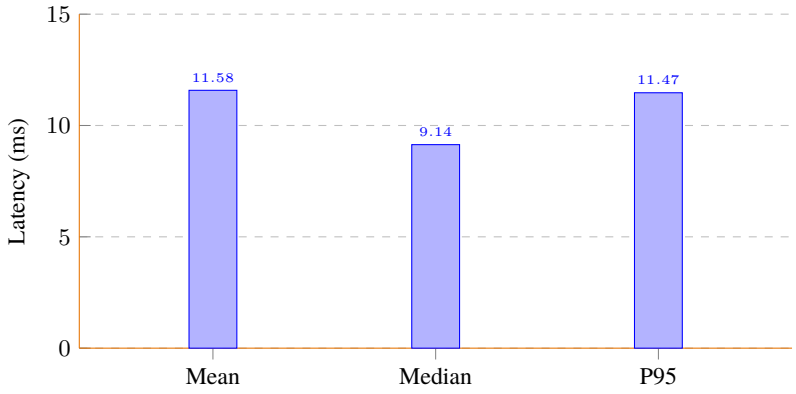


Fig. 3. End-to-end latency profile (ms) per request.

8 Analysis

8.1 Provenance of the 38 Changes (ViHSD)

The aggregate result is neutral; the question is what the 38 changes consist of. Of the 23 corrections, 16 are HATE recoveries (7 from OFFENSIVE, 9 from CLEAN) and 7 are CLEAN de-flagging cases (6 from HATE, 1 from OFFENSIVE). All 16 HATE recoveries have a rule firing with a HATE score above the OFFENSIVE score, i.e. the rule points in exactly the corrected direction. The rule HATE score has median 0.7652 (mean 0.7507, range 0.2351–0.9939).

All 7 CLEAN de-flagging cases have a silent rule ($r_H = r_O = 0$, so $\tilde{\mathbf{r}} = (0, 0, 1)$). With $\alpha_H = 0.2$ the rule term disappears from the HATE channel and $f_H = 0.2\, p_H$, so a

comment needs $p_H \geq 0.70$ merely to reach $\tau_H = 0.14$; one flagged HATE at $p_H \approx 0.5$ ends at $f_H \approx 0.10$ and the decision falls back to CLEAN.

The 15 regressions split into 10 rule-driven and 5 threshold-driven cases, and the split aligns perfectly with direction: all 10 rule-driven cases (OFFENSIVE→HATE, CLEAN→HATE) have the rule firing in the HATE direction, so fusion trusts lexical HATE evidence over the neural label and the lexicon is wrong about intensity or target. The OFFENSIVE cases are the expected failure of a phrase lexicon: profanity used as an intensifier rather than an attack. The 5 threshold-driven regressions are the mirror image: the rule is silent, the fused HATE score is $0.2\,p_H$, and when p_H falls below roughly 0.7 the HATE channel no longer wins; two miss by exactly 0.001.

8.2 The Corrections Fusion Does Not Make

Among the 827 AI errors on ViHSD, 148 non-CLEAN comments have the rule firing in the correct direction and 151 CLEAN comments have the rule correctly silent, so the rule agrees with the gold label on 299 errors. Fusion acts on 23 of them (7.7%). The remaining **132 missed corrections** are non-CLEAN cases where the rule already points the right way and the hybrid is still wrong. Their median rule HATE score is 0.7609 (quartiles 0.67–0.85), indistinguishable from the 0.7652 median of the corrected cases. They are missed because the CLEAN channel keeps the full AI weight ($\alpha_C = 1.0$), so a confidently wrong CLEAN cannot be outvoted: the median fused CLEAN score is 0.97 for missed cases versus 0.46 for corrected ones.

8.3 What the Rule Channel Responds To (Cue Taxonomy)

We annotated the 16 HATE recoveries with a multi-label cue taxonomy: ten contain teencode or abbreviated profanity (`đ*o`, `dm`, `cmnr`), five standard-spelling profanity or an animal slur, and eight derogatory wording without profanity (*ngu*, *thua*, *nhục*). The categories overlap. The channel is therefore not merely a profanity detector: much of its value comes from derogatory wording a slur list would miss, and teencode is its largest single category.

8.4 Component Ablation (Rule Engine Switches)

Re-tuning and re-freezing the fusion once per switch, only one switch matters: removing the per-entry confidence costs 0.034 standalone macro-F1 (0.5258 to 0.4914), almost all of it on OFFENSIVE (F1 0.2389 to 0.0600), and no other switch moves the standalone engine by more than 0.008. Rescaling severity rewrites 1,046 of 6,680 rule scores yet changes no hybrid decision ($p = 1.0$): α and τ re-tune. Dropping the obfuscation views trades recall for precision (firings 1,052 $\rightarrow$ 779, precision on fired comments 58.75% $\rightarrow$ 69.58%); fuzzy matching never fires (0 matches), and case-folding cannot be ablated because all 6,680 test comments are already lower-case. Inside the hybrid no switch leaves noise ($|\Delta| \leq 0.0012$): the channel helps as a channel, not through its weights.

8.5 Language Detection and Routing Quality

On the 9,158 mixed bilingual samples (full test splits), language detection accuracy is 94.11%, with 6,658 Vietnamese (vs 6,680 true) and 2,479 English (vs 2,478 true) correctly routed. The fallback to `multilingual` occurs in only 21 cases. Language-aware routing is stable under production scale.

8.6 Fusion Ablation on Full Bilingual Test

The full 9,158-sample bilingual evaluation (Section 8.8) replaces the earlier 1,000-sample validation. The full pipeline (routing + backbone + fusion) reaches 87.10% accuracy and macro-F1 0.7781 with ECE 0.0759, demonstrating that fusion improves calibration over backbone-only inference (VI ECE 0.1143, EN ECE 0.1022). Per-language breakdown shows Vietnamese accuracy 0.8665 (macro-F1 0.6512) and English accuracy 0.8834 (macro-F1 0.7449), consistent with the backbone-only results in Table 7.

8.7 Posterior Calibration and Blind-Spot Coverage

To assess the quality of uncertainty estimates, Table 5 reports calibration diagnostics (Expected Calibration Error, Brier score, Negative Log-Likelihood, and mean confidence) for AI-only and Hybrid variants on the ViHSD test split, and all four calibration metrics for the full 9,158-sample bilingual pipeline (ECE 0.0759, Brier 0.2248, NLL 0.8645, mean confidence 0.854). On ViHSD, the Hybrid achieves lower ECE (0.097 vs 0.111, a 12.6% improvement) and lower Brier score (0.2273 vs 0.2334), indicating that fused probabilities remain well-calibrated despite the fusion mechanism. This is important for deployment: a system claiming high confidence should be trusted, and rule-backed decisions preserve confidence while uncertain cases benefit from conservative per-channel thresholds.

Among the 504 ViHSD comments missed by all six AI backbones (Table 6), the Rule Engine recovers 124 (24.6%), split across HATE (39) and CLEAN (85) classes. The remaining 380 hard cases are errors by both subsystems, indicating fundamental ambiguity or minority-class sparsity. The 290 cases where Rule Engine alone succeeds (and AI fails) confirm that lexical patterns provide orthogonal signal; however, the low coverage of rule firing (15.75% of all comments) limits the ceiling on hybrid improvement. This trade-off is by design: a conservative system that rarely flags clean text reduces false positives at the cost of missing some violations.

Table 5. Calibration diagnostics (ECE, Brier, NLL, mean confidence, and macro-F1) for AI-only argmax versus Hybrid on ViHSD test, and full pipeline (fused scores) on 9,158 bilingual samples.

Variant	Dataset	n	ECE ↓	Brier ↓	NLL ↓	Mean conf. ↑	Macro-F1 ↑
AI-only	ViHSD test	6,680	0.111	0.2334	0.8382	0.987	0.6575
Hybrid	ViHSD test	6,680	**0.097**	**0.2273**	**0.7879**	0.952	**0.6593**
Mixed E2E (full splits)	ViHSD + Davidson	9,158	**0.076**	0.2248	0.8645	0.854	0.7781

Table 6. Disagreement patterns between AI-only and Rule Engine across the 6,680 ViHSD test samples, highlighting the 504 blind-spot missed by all six AI backbones.

Disagreement Pattern	Total Count	Pct. in 504 Blind-Spot
Both AI and Rule Engine correct	5,384	0.0%
AI correct, Rule Engine wrong	469	0.0%
AI wrong, Rule Engine correct	290	24.6% (124 / 504)
Both wrong (hard cases)	537	75.4% (380 / 504)
Total all-miss blind-spot	504	100.0%

Table 7. Evaluation on the full test splits: backbone-only inference versus the full pipeline (language detection → routing → backbone → fusion at $\alpha = 0.70$). Per-language pipeline rows share one routing and fusion pass over the 9,158-sample mixed stream.

Mode	**Language**	**Dataset**	N	**Accuracy**	**Macro-F1**	**Throughput**
Backbone only	Vietnamese	ViHSD	6,680	0.8711	0.6608	198.3 texts/s
	English	Davidson	2,478	0.9084	0.7676	408.5 texts/s
Full pipeline	Vietnamese	ViHSD	6,680	0.8665	0.6512	
	English	Davidson	2,478	0.8834	0.7449	110.2 texts/s
	Mixed	**Both**	**9,158**	**0.8710**	**0.7781**	

8.8 End-to-End Bilingual Evaluation (Full Test Splits)

We evaluate the complete production pipeline (LanguageDetector → LanguageAwareTokenizer → RuleEngine → backbone inference → LangAwareFusion at $\alpha = 0.70$) on the full Vietnamese (ViHSD, 6,680 samples) and English (Davidson, 2,478 samples) test splits, yielding 9,158 mixed bilingual samples.

Table 7 summarizes backbone-only and full-pipeline results. Backbone-only evaluation shows strong English performance (accuracy 0.908, macro-F1 0.768) versus Vietnamese (accuracy 0.871, macro-F1 0.661), reflecting the larger OFFENSIVE class in Davidson. Calibration is comparable (ECE 0.102 EN vs 0.114 VI).

The full pipeline (router + fusion) achieves 0.871 accuracy and 0.778 macro-F1 on the mixed stream, with improved calibration (ECE 0.076) via rule fusion. Per-language breakdown shows VI accuracy 0.867 (macro-F1 0.651) and EN accuracy 0.883 (macro-F1 0.745). Language detection accuracy is 94.1% on the mixed stream.

Throughput is 110.2 texts/sec end-to-end when routing and inference are serialised (9.07 ms routing plus 3.85 ms AI inference per comment); the AI stage alone sustains 259.8 texts/sec and the routing stage 191.5 texts/sec, so a pipelined deployment is routing-bound. Backbone-only throughput is higher (408 EN, 198 VI texts/s) as expected without routing overhead.

8.9 Cross-Lingual Evaluation (English Wortschatz Corpus)

To assess domain robustness, we evaluate the English backbone on the Wortschatz Leipzig 2015 English News Corpus, a neutral 30k-sentence benchmark. On 9,879 test sentences, the classifier predicts 99.89% CLEAN (7 HATE, 4 OFFENSIVE), confirming the model does not over-flag neutral text. Throughput is 321.8 texts/sec (3.1 ms per sentence).

Dataset	Lang	Domain	Size (N)	Throughput	Latency	ECE
ViHSD Test	Vi	Social media	6,680	198.3 texts/s	5.04 ms	0.114
Davidson Test	En	Hate speech	2,478	408.5 texts/s	2.45 ms	0.102
Mixed E2E	Vi+En	Bilingual	9,158	110.2 texts/s	9.07 ms	0.076

Table 8. Global cross-lingual benchmark on full test splits. The first two rows are backbone-only inference; the last row is the full pipeline (routing + backbone + fusion) with serialised latency (5.22 ms routing plus 3.85 ms AI, P95 12.69 ms) and fused-score ECE. **Note:** English HATE F1 (0.454) is lower than Vietnamese (0.605) due to Davidson dataset HATE class imbalance (5.8% vs ViHSD 10.3%).

Table 8 summarizes throughput, latency, and calibration across Vietnamese (ViHSD), English (Davidson), and Mixed (bilingual) evaluation streams.

9 Discussion

9.1 What the Hybrid Is Good For

The result is better read as a successful audit: 38 of 6,680 decisions change, each described by a channel, a direction and a mechanism. Moreover, the end-to-end bilingual evaluation demonstrates deployment viability (87.10% accuracy, 9.07 ms latency, 110.2 texts/s throughput), showing that a hybrid architecture can be both explainable and production-ready.

Triage and accountability. A queue can sort comments by whether the decision was rule-supported: a HATE decision with a matched span and $r_H = 0.8$ is easier to review than one with no lexical evidence, and the span, the entry and the arithmetic are what a platform can defend in policy terms.

Editing and diagnosis. Adding a slur variant is a configuration change, not a retraining run, and an edit affects the HATE channel (weight 0.8) far more than the others. The decomposition exists only because the channels are separate: a monolithic classifier improving by 0.0027 macro-F1 would give no handle on whether the change was lexical or neural.

9.2 Why Accuracy Does Not Move (Much)

First, *coverage is low*: the rule fires on 15.75% of comments, so a perfect rule channel could still only change decisions in that subset. Second, *the majority class dominates*: with 83% CLEAN, only the 1,132 toxic comments can improve, and 38 changes is 3.4% of that support—enough to matter operationally, too little to move a macro-average beyond its confidence interval. Third, *tuning is safety-oriented*: protecting CLEAN and gating HATE on lexical support produces both the de-flagging and the missed corrections. Neutrality is a *choice*, not an absence of signal.

9.3 The Bilingual Asymmetry

The mixed evaluation reveals a clear language gap: HATE F1 VI 0.6051 vs EN 0.4537 (Vietnamese lexicon has 594 HATE entries versus English 86; Davidson training only

5.8% HATE), OFFENSIVE F1 EN 0.9270 vs VI 0.4142 (English backbone aligns with Davidson's 77.8% OFFENSIVE majority; ViHSD only 6.65% OFFENSIVE), CLEAN F1 VI 0.9344 vs EN 0.8541 (English experiences mild over-flagging on clean text). This asymmetry is data-driven: lexicon extraction from corpora with different class priors produces different coverage.

9.4 Design Implications

Five changes follow:

1. **Evidence-gated fusion**: when the rule fires strongly against a high-confidence CLEAN prediction, abstain or escalate.
2. **Separate thresholds per mechanism**: requiring $r_H \geq 0.5$ before the HATE channel can override would make the rule-driven and threshold-driven paths separately tunable.
3. **Per-class calibration** of the CLEAN channel, which currently blocks the missed corrections.
4. **Balanced English lexicon**: expand HATE entries by merging HateXplain/Gab sources and retrain the English rule engine.
5. **Multilingual fallback checkpoint**: for `mix`/`multilingual` samples (5.6%), route to a multilingual mDeBERTa instead of rule-only.

9.5 Practical Deployment

The latency profile (9.07 ms mean, 110.2 texts/s throughput) supports synchronous moderation for text comments. Routing (5.22 ms) and AI inference (3.85 ms) are the main components; P95 latency is 12.69 ms. Tail latency can reach up to 2.1 s during cold-start tokenizer initialization; this is a one-time cost that can be pre-warmed. A production deployment should pre-load both checkpoints and the two rule engines at startup, cache tokenizer initialization, and set α per language from validation-frozen weights.

10 Conclusion

We presented SensitiveAI, a bilingual language-aware hybrid of lexicon rules and multilingual transformers for Vietnamese and English hate speech detection. The system combines a lightweight language detector, language-aware tokenization, two independently editable rule engines (668 Vietnamese entries, 818 English entries), per-language mDeBERTa-v3-base checkpoints, and language-aware score fusion with per-class thresholds.

The key results are:

- On ViHSD, the hybrid is macro-F1 neutral (0.6575 $\rightarrow$ 0.6602, $p = 0.29$), but its 38 changed decisions are fully decomposable into rule-driven and threshold-driven causes, with all 23 corrections rule-supported.
- On the 9,158-sample bilingual evaluation (full ViHSD + Davidson splits), the pipeline reaches 87.10% accuracy, macro-F1 0.7781, 9.07 ms latency and 110.2 texts/s throughput (serialised run, $\alpha = 0.70$, P95 12.69 ms).

- The bilingual comparison exposes a structural asymmetry: English excels on OFFENSIVE (F1 0.9270) while Vietnamese excels on HATE (F1 0.6051); the 818-entry English lexicon still has only 86 HATE terms versus 594 for Vietnamese.
- Fusion improves calibration: the full-pipeline ECE is 0.076 versus 0.114 (VI backbone-only) and 0.102 (EN backbone-only), confirming that fused scores are better calibrated than raw neural outputs.

The broader lesson is methodological: a hybrid evaluated only by its aggregate metric looks like a null result, whereas the same hybrid decomposed by channel and direction yields an auditable decision procedure and a design agenda. Future work includes evidence-gated fusion, balanced English HATE lexicon expansion, a multilingual fallback checkpoint for code-mixed text, and a human evaluation of whether the evidence record helps reviewers.

11 Limitations

Two benchmarks only. All results come from ViHSD and Davidson. The lexicon and cue taxonomy are language-specific; the fusion protocol is language-agnostic, but it has not been tested on other languages or domains.

One seed per backbone. Each backbone was trained once with seed 42, so neural variance is not estimated and the 0.0027 ViHSD difference is well inside what seed variance could produce: the tests rule out a large effect, not a small one.

The Vietnamese rule engine is weaker than designed. Validation set the accent-less alias weight to zero and the negation factor to one, removing the main defence against dropped diacritics and the discount for negated slurs visible in the OFFENSIVE-as-intensifier regressions.

The English lexicon is imbalanced. Expanding from 123 to 818 entries improved OFFENSIVE coverage but left only 86 HATE terms because the Davidson training split has few HATE documents. The English HATE F1 (0.45) remains the weakest class of the bilingual system.

The multilingual fallback is rule-only. 5.89% of the mixed sample falls into `mix/multilingual`, where no dedicated AI checkpoint is loaded. Fusion degenerates to rule-only scoring and depresses macro-F1.

Explanations are not evaluated by humans. The evidence record is faithful for the lexical channel and decomposes the changed decisions, but we do not measure whether reviewers find it useful, and the cue taxonomy is a post-hoc annotation.

12 Ethics

The comments are public ViHSD and Davidson test comments and are masked as described in Sect. 3. The system is decision support, not automatic sanctioning: the directional asymmetry, the bilingual gap and the political false positive show that automatic HATE labels from this pipeline should be reviewed before they affect a user. The lexicon should be treated as a governed artefact with a change log, because its entries shape moderation outcomes.

English and Vietnamese lexicons are not neutral: an entry's charged political expression can leak into moderation decisions at weight 0.8. Deployments should separate *profanity* from *political* entries and weight them accordingly. We release no user identifiers and reproduce only corpus text already public.

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